Development and Implementation of an Interactive Web-Based Diabetes Risk Assessment

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Independent Project

Author Note

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Abstract

This paper describes the design, development, and implementation of an interactive web-based diabetes risk assessment tool. The application was built to translate multiple clinical and lifestyle risk factors into an accessible, visual risk classification with personalized recommendations. The system was developed using FastAPI (backend), HTML/CSS/JavaScript (frontend) and matplotlib (visualization). This methodology employed a rule-based risk scoring model with eight weighted risk components: age (0.20), BMI (0.25), family history (0.15), exercise frequency (0.15), diet quality (0.10), blood pressure (0.08), fasting blood glucose (0.05), and lifestyle composite (0.02). The tool supports both full and quick assessment modes, renders interactive risk factor charts, and stores assessment history in browser local storage. Testing confirmed functionality across key user workflows and responsive design on mobile and desktop devices. Results demonstrate a prototype suitable for educational use and decision-support exploration. Limitations include the absence of external dataset validation and the lack of clinical trial evidence. Future work should include integration with empirical datasets and performance validation against a gold-standard criteria.

Keywords: Diabetes Risk Assessment, web application, rule-based scoring, risk visualization, mobile-responsive design, health informatics, Quick Assessment

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The rise of diabetes, especially type 2 diabetes, has created demand for scalable, accessible risk assessment tools (Diabetes, 2023). Current web-based screening platforms, such as those offered by the American Diabetes Association and Centers for Disease Control and Prevention, provide categorical risk results but often lack visualization, personalized recommendations, and interactive history tracking (CDC, 2023). This project addresses that gap by developing a responsive, interactive web application that combines a rule-based risk model with modern user interface design and client-side data persistence. The tool is designed to be deployed on cloud platforms with HTTPS support, making it freely accessible to users worldwide.

Methods

System Design and Architecture

The system employs a client-server architecture with a Python backend (FastAPI) and a responsive web frontend (HTML, CSS, JavaScript). User inputs are validated on the client side and submitted to the backend via HTTP POST requests to one of two routes: '/submit' (full assessment) or '/quick_submit' (quick assessment). The backend creates a 'PatientProfile' dataclass, computes risk scores using the 'DiabetesRiskCalculator' class, renders base64-encoded PNG charts using matplotlib, and returns a complete HTML response with results and visualization. Assessment data is then automatically saved to a local storage via JavaScript.

Risk Scoring Model

Participants and Input Variables

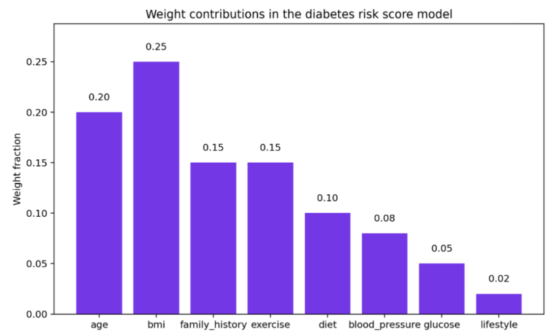
The model accepts data on 10 health and lifestyle dimensions: age (years), BMI (kg/m²), family history of diabetes (binary) exercise frequency (days per week), diet quality (categorical), systolic blood pressure (mmHg), fasting blood glucose (mg/dL), stress level (scale), sleep hours (average per night), and current medications for hypertension or cholesterol (binary). These variables were selected based on established diabetes studies and clinical practice guidelines.

Assessments and Measures

Feature-Level Risk Scoring. Each input variable is converted to a normalized risk score (0.0–1.0) using predefined threshold bins. Age-based scoring reflects increased diabetes incidence with advancing age: individuals under 30 receive a score of 0.1, while those 60 and older receive 0. 9. BMI thresholds align with WHO and clinical guidelines, with normal weight (18.5–24.9) scored at 0.2 and severe obesity (35+) at 1.0. Exercise frequency follows a dose-response relationship: 5–7 days per week yields 0.1 (low risk), while no behavior (0 days) yields 1.0. Diet quality is categorical: good (0.2), fair (0.6), poor (1.0). Blood pressure thresholds classify normal (≤120 mmHg: 0.2), elevated (121–140: 0. 4), and hypertensive (>140: 0.8). Fasting blood glucose uses clinical cutoffs: normal (≤100: 0.2), prediabetic range (101–126: 0.6), and diabetic range (>126: 0.9). Family history is binary (yes: 0.7, no: 0.1). The lifestyle composite score integrates stress and sleep: `0.5 × (stress / 10) + 0.5 × (| sleep − 7| / 7)`, reflecting both psychological stress and sleep deviation from the recommended 7 hours.

Weight Aggregation. The overall risk score is computed as a weighted sum of the eight risk components: Overall Risk Score = Σ(Feature Score × Weight), where weights are: age (0.20), BMI (0. 25), family history (0.15), exercise (0. 15), diet (0.10), blood pressure (0.08), glucose (0.05), and lifestyle (0.02) (seen in Figure 1). These weights reflect assumptions about the relative importance of each factor in diabetes risk and were set primarily on clinical reasoning rather than empirical model fitting.

Figure 1- *Weight Contributions of risk factors in the assessment model*
This bar chart displays the relative weight assigned to each factor within the DiabetesRiskCalc scoring algorithm. The figure uses original data from the application source code



Risk Classification. The computed score is mapped to three categorical risk levels: HIGH RISK (score ≥ 0.70), MODERATE RISK (score ≥ 0. 50), and LOW RISK (score < 0.50).

Quick Assessment Mode

For users preferring faster input, the quick assessment mode collects only six variables: age, BMI, family history, exercise frequency, diet quality, and name. Default values are assumed for unmeasured clinical variables: systolic blood pressure = 120 mmHg, fasting glucose = 95 mg/dL, stress level = 5, sleep = 7 hours, and medications = no. These defaults approximate population medians and allow rapid risk screening in the absence of complete clinical data.

Figure 1- *Quick Assessment Mode Form*
The quick assessment requires only six inputs—age, BMI, family history, exercise, diet quality, and an optional name—reducing completion time to about 1–2 minutes. Missing factors are assigned default values to streamline the process.

Visualization and User Interface

Risk factor scores are rendered as a horizontal bar chart using matplotlib's Agg backend (for server compatibility), with color-coding: red for high risk (>0. 7), orange for moderate (0.4–0.7), and green for low (<0.4). An additional gauge-style visualization displays the overall risk score. All charts are converted to base64-encoded PNG images and embedded, eliminating server storage and simplifying deployment.The user interface employs responsive CSS grid layouts, color-coded risk badges, helper text for each input, and toggle buttons for switching between full and quick modes. The design uses a white-to-blue gradient background with light blue card containers, optimized for mobile, tablet, and desktop viewports.

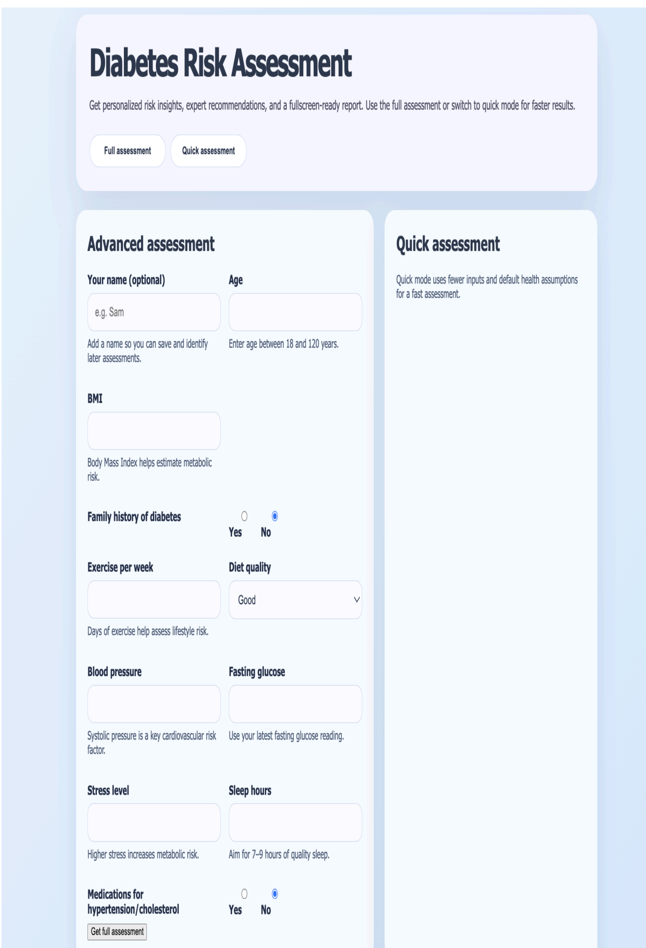


Figure 1- *Homepage with Full Assessment Form*
The landing page features an advanced assessment form with ten input fields, helper text, a responsive grid layout, and toggle buttons for switching between full and quick modes. A blue gradient background and white card design provide a clean, mobile-friendly interface.

Implementation Environment

The backend was implemented in Python 3 using FastAPI version 0.104.1 and Uvicorn version 0.24.0. Charting was performed with matplotlib version 3.8.2 configured with the Agg backend. Front-end code uses vanilla HTML5, CSS3 with flexbox and CSS grid, and ES6 JavaScript. Browser local storage (`localStorage` API) persists assessment history as JSON. The application is designed for deployment on cloud platforms (Render, Railway, Heroku) via a Procfile that specifies the startup command: `uvicorn main:app --host 0.0.0. 0 --port $PORT`.

Testing and Validation

The application was tested by submitting sample user inputs across both full and quick assessment pathways. Form validation logic was verified to ensure inputs remained within specified ranges. Responsive design was checked by resizing the browser window and testing on simulated mobile viewports. Chart generation was verified to confirm base64 encoding and proper display within HTML responses. Error handling was tested by submitting invalid data and confirming graceful error page display.

Results

Functional Outcomes

Outcome #1: Risk Computation and Categorization

The system successfully computes weighted risk scores and assigns appropriate risk categories. Sample test cases confirmed the mathematical correctness of the scoring pipeline and the proper classification of users into low, moderate, and high risk groups. For example, a 45-year-old with BMI 32, sedentary lifestyle, and poor diet quality was correctly classified as

MODERATE RISK (seen in Figure 4-5), while a 28-year-old with normal BMI and active lifestyle was classified as LOW RISK (seen in Figure 2-3).

Figure 2- *Low-risk profile factor score breakdown*
Factor scores are calculated from age, BMI, family history, exercise, diet, blood pressure, blood glucose, and lifestyle.

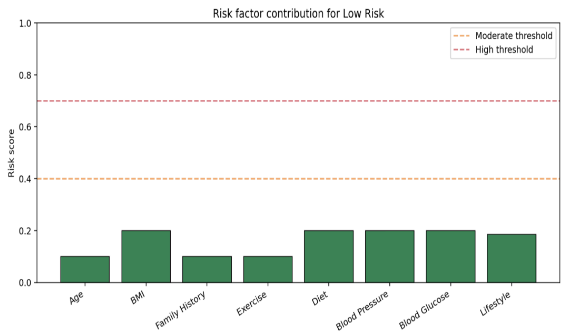


Figure 3*- Low-risk profile overall risk gauge*
The gauge depicts a combined weighted score on a 0-1 scale, with the filled bar showing the subject's overall risk .Thresholds follow the model's low/moderate/high boundaries.

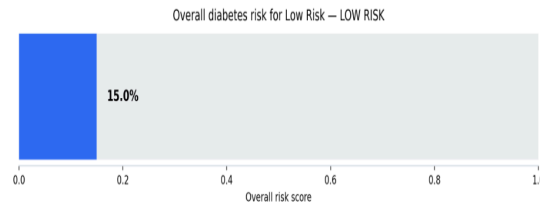


Figure 4- *Moderate-risk profile factor score breakdown*
The moderate-risk example includes elevated BMI and family history, with exercise and diet contributing to the overall score.

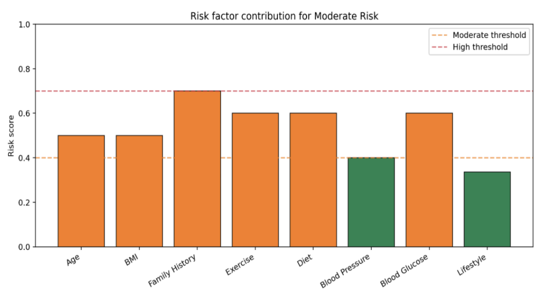
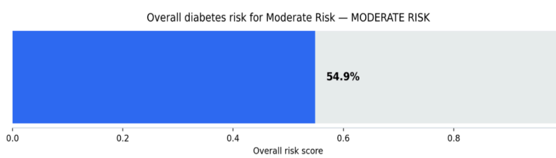


Figure 5*- Moderate-risk profile overall risk gauge*
The gauge illustrates moderate risk where the overall weighted score falls between the defined moderate and high thresholds. This figure is derived from the model's internal scoring logic



Outcome #2: Visualization and User Experience

Charts are generated and embedded within response HTML without errors. The bar chart of risk factors displays with appropriate color coding and legend. The gauge visualization renders clearly and labels the overall risk category. The responsive layout adapts correctly to mobile widths (320–480 px), tablet widths (768–1024 px), and desktop widths (1200+ px), with form inputs, buttons, and text remaining legible and functional across all viewports.

Data Persistence

Assessment history is successfully written to and read from browser local storage. Users can view a chronological history of past assessments (stored as name, age, risk category, and date). The history table updates in real time upon new submissions. The clear history function successfully removes all stored records (Seen in Figure 6).

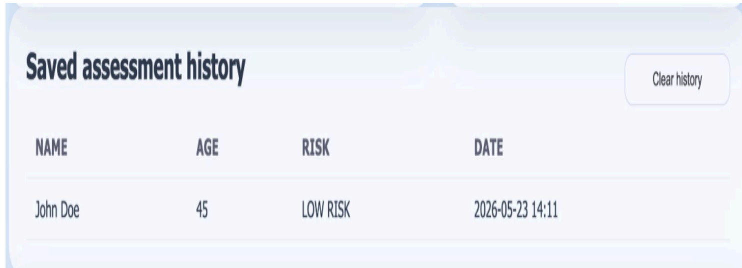


Figure 6- *Assessment history tracking interface*
The user interface displays name, age, risk level and the date of which the assessment occurred.The assessment can be cleared through the button, but cannot be retrieved

Design and Usability

The final user interface presents information hierarchically, with the overall risk category prominently displayed, followed by a detailed breakdown of input values, top risk factors, and actionable recommendations. The quick assessment mode reduces the typical input time from 3–4 minutes to approximately 1–2 minutes by eliminating clinical measurements with safe defaults. No formal usability testing with external users was conducted; however, the interface navigation and feature discoverability appear intuitive based on developer review.

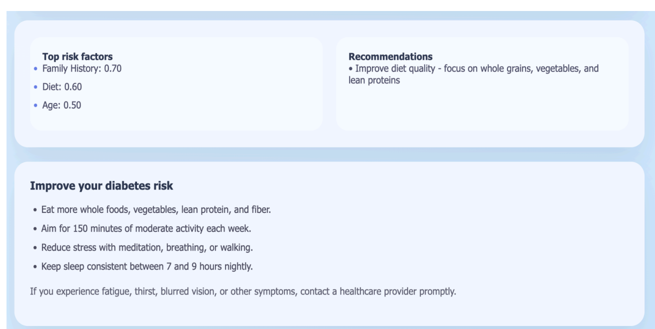


Figure 6- *Results Page Showing Risk Assessment Output*
The results page displays diabetes risk factors, personalized recommendations and health tips based on the user's assessment. A clean card-based layout and blue-and-white design makes the information easy to read and navigate.

Discussion

Achievement of Project Goals

The project successfully created a functional, deployable web application for diabetes risk assessment. The system meets the primary objectives of translating multiple risk factors into a clear classification, providing visual feedback, offering personalized guidance, and enabling history tracking. The responsive design ensures usability across modern devices, and the rule-based model is transparent and interpretable.

Technical Strengths

The architecture is lightweight and scalable, suitable for cloud deployment. The use of base64-encoded chart images eliminates server-side file storage overhead. The separation of full and quick assessment modes accommodates users with varying time availability and clinical data availability. Browser-based history storage reduces server load and respects user privacy by keeping assessment data local.

Limits and Constraints

The primary limitation is the absence of external dataset validation. The rule-based thresholds and weights were selected using clinical epidemiological reasoning but have not been tested against real-world cohorts with confirmed diabetes diagnoses. The model lacks calibration: a score of 0.65 does not necessarily represent a 65% probability of developing diabetes. The application should be regarded as an educational and exploratory tool, not a clinical diagnostic instrument. No formal comparison with existing risk calculators (e.g., ADA, CDC) was performed. The quick assessment mode uses fixed defaults that may not reflect individual variation in unmeasured factors.

Future Research Directions

To advance this work toward a validated clinical tool, the following steps are recommended:

1. Acquire a representative diabetes dataset (e.g., Pima Indians Diabetes Database, NHANES, Framingham Heart Study, UK Biobank) and train a data-driven model (logistic regression, random forest, or gradient boosting) to predict diabetes incidence or presence.

2. Perform model validation: compute AUC, accuracy, sensitivity, specificity, and calibration metrics. Compare the data-driven model against the current rule-based approach.

3. Validate thresholds using clinical guidelines and empirical data from large cohorts. Adjust weights and cutoffs to optimize predictive performance.

4. Conduct a usability study with a diverse user population to gather feedback on interface design, clarity of results, and actionability of recommendations.

5. Implement user accounts and cloud-based data storage to enable longitudinal tracking and risk trend analysis.

6. Integrate real-time clinical data APIs (e.g., FHIR-compliant EHR systems) to populate inputs from medical records.

Implications

If validated, such a tool could improve diabetes screening and awareness in community settings and support clinical decision-making in primary care. The transparent rule-based model is particularly valuable for educational use, as users can understand which factors drive their risk and how lifestyle or clinical interventions might modify scores. The responsive design and accessibility (HTTPS deployment) positions it as a scalable solution for global health awareness campaigns.

Conclusion

This project demonstrates the feasibility of translating a rule-based diabetes risk model into an interactive, visually engaging web application. The system successfully integrates multiple data inputs, applies a transparent weighted scoring algorithm, generates informative visualizations, and persists user data across sessions. The responsive design and quick assessment mode enhance accessibility and usability. While the prototype is suitable for educational and exploratory purposes, rigorous validation against external datasets and clinical outcomes is necessary before clinical deployment. Future work should focus on model validation, usability testing, and integration of empirical data to transform this prototype into an evidence-based, clinically trusted risk assessment tool.

Appendices

Appendix A: Core Risk Calculation Function

```python
def calculate_overall_risk(self, profile:
PatientProfile) -> Tuple[float, Dict]:
    """Calculate comprehensive risk score"""
    scores = {
        "Age": self.calculate_age_risk
(profile.age),
        "BMI": self.calculate_bmi_risk
(profile.bmi),
        "Family History": 0.7 if profile.
family_history else 0.1,
        "Exercise": self.
calculate_exercise_risk(profile.
exercise_frequency),
        "Diet": self.calculate_diet_risk
(profile.diet_quality),
        "Blood Pressure": 0.8 if profile.
blood_pressure_systolic > 140 else 0.
4 if profile.blood_pressure_systolic
> 120 else 0.2,
        "Blood Glucose": 0.9 if profile.
blood_glucose_fasting > 126 else 0.6
if profile.blood_glucose_fasting >
100 else 0.2,
        "Lifestyle": (profile.stress_level /
10) * 0.5 + (abs(profile.sleep_hours
- 7) / 7) * 0.5
    }

    factor_weight_keys = {
        "Age": "age",
        "BMI": "bmi",
        "Family History": "family_history",
        "Exercise": "exercise",
        "Diet": "diet",
        "Blood Pressure": "blood_pressure",
        "Blood Glucose": "glucose",
        "Lifestyle": "lifestyle"
    }
    weighted_score = sum(
        scores[factor] * self.WEIGHTS
[factor_weight_keys[factor]]
        for factor in scores
    )

    return weighted_score, scores
```

Appendix B: Partial HTML Form Structure

The full assessment form collects the following fields in a responsive two-column grid:

- Age (required, range 18–120)
- BMI (required, range 10–60, decimal step 0. 1)
- Family history (radio: yes/no)
- Exercise frequency (required, range 0–7)
- Diet quality (dropdown: good/fair/poor)
- Blood pressure systolic (required, range 80–200)
- Fasting blood glucose (required, range 60–300)
- Stress level (required, range 1–10)
- Sleep hours (required, range 3–12, decimal step 0.1)
- Medications (radio: yes/no)

Appendix C: Deployment Procfile

```
web: uvicorn main:app --host 0.0.0.0 --port
$PORT
```

This command instructs cloud platforms (Render, Railway, Heroku) to start the Uvicorn server on the platform-provided port, enabling web traffic over HTTPS.

Appendix D: Risk Threshold Summary Table

Risk Level	Score Range	Interpretation
Low	< 0.50	Minimal diabetes risk; maintain current lifestyle
Moderate	0.50–0.69	Elevated risk; consider lifestyle modifications
High	≥ 0.70	Substantial risk; consult healthcare provider